

FLEX-UF: Spatially Adaptive Early Exit in DCVC-UF

Anonymous CVPR submission

Paper ID DRAFT

Abstract

The benefit of another decoder layer varies across an image, yet DCVC-UF applies the same synthesis depth everywhere. We introduce FLEX-UF, a spatial early-exit extension of its intra model. A shared coded representation and synthesis prefix feed tiles that stop at different depths. Exit adapters assemble their features for one full-frame repair and reconstruction head; only surviving tiles execute later blocks. We evaluate two distinct sources of efficiency: reducing average depth and choosing where to retain it. Across 53 intra frames at five quality indices, learned routing saves 27.97% of modelled synthesis multiply-accumulates at a nominal 0.1 dB YCbCr-MSE target under per-source calibration. Of this saving, 2.54 percentage points exceed ordered dithering, which mixes depths without predicting their spatial value. Retrospective selection under a common measured-loss cap retains a 2.93-point margin over all 265 pairs. This margin contracts as the shallowest exit saturates. Fixing controls on other sequences removes per-source calibration, but a QP32 reconstruction replay does not yet resolve the paired routing advantage. These results identify where adaptive depth can add value while separating arithmetic savings from unmeasured codec latency. An ongoing independent-depth study extends the same allocation question from shared exits to codec selection.

1. Introduction

Consider two regions decoded from the same image. In one, another pair of synthesis blocks improves reconstruction; in the other, it changes little. A fixed-depth decoder pays for both. Efficient operators make this work cheaper, but leave the allocation unchanged. Spatial early exit asks a complementary question: *where is further decoding worth its cost?*

We investigate this question in the intra model supplied with DCVC-UF [6]. Its synthesis transform contains twelve constant-width blocks between upsampling and reconstruction, providing a concrete axis along which to vary capacity. We turn this fixed stack into an image-dependent allocation

of depth. Following the incremental-benefit view of adaptive patch exiting [9], a useful decision reflects what additional processing can recover, rather than texture alone. In a codec, that benefit also depends on the information preserved at the chosen rate: a visually complex region need not justify the deepest reconstruction.

There are two natural ways to make this decision (Fig. 1). *Across models*, a selector assigns each patch to a codec of suitable capacity, following the patch-to-network principle of ClassSR [5]. *Within a model*, different patches leave a shared synthesis stack at different depths. The distinction is where the commitment is made. An independent codec is chosen before encoding and can learn its own representation; an exit is chosen after one representation has been recovered and reuses its synthesis prefix. FLEX-UF implements and evaluates the latter. The six-depth bank is a complementary study of how specialisation trades against this reuse.

Inside a codec, early exit requires more than attaching intermediate outputs. Features at different depths must support a common reconstruction; conditional tiles lose some spatial context; and choosing an exit can itself require expensive candidate decoding. FLEX-UF addresses execution with a full-frame stem, a shrinking batch of active tiles, point-wise exit adapters, and a shared repair and reconstruction head. Exited tiles leave the active batch, making the conditional path an actual reduction in executed blocks. The frozen analysis and entropy path supplies one common latent; its payload can support different stopping maps without re-encoding the image.

We separate the value of *cheaper depth* from the value of *content-dependent placement*. Uniform depth tests global truncation. Ordered dithering permits mixtures of depths without predicting where extra synthesis is useful. Under source-calibrated control, learned routing improves on this stronger baseline by 2.54 percentage points of synthesis MAC saving at a nominal 0.1 dB target, above dithering's 25.43% reduction. That increment shrinks near the shallowest-exit ceiling. Adaptation has most room to help when extra depth still matters but does not matter equally everywhere. A common delivered-loss analysis retains this pattern. A separate QP32 replay fixes controls on other

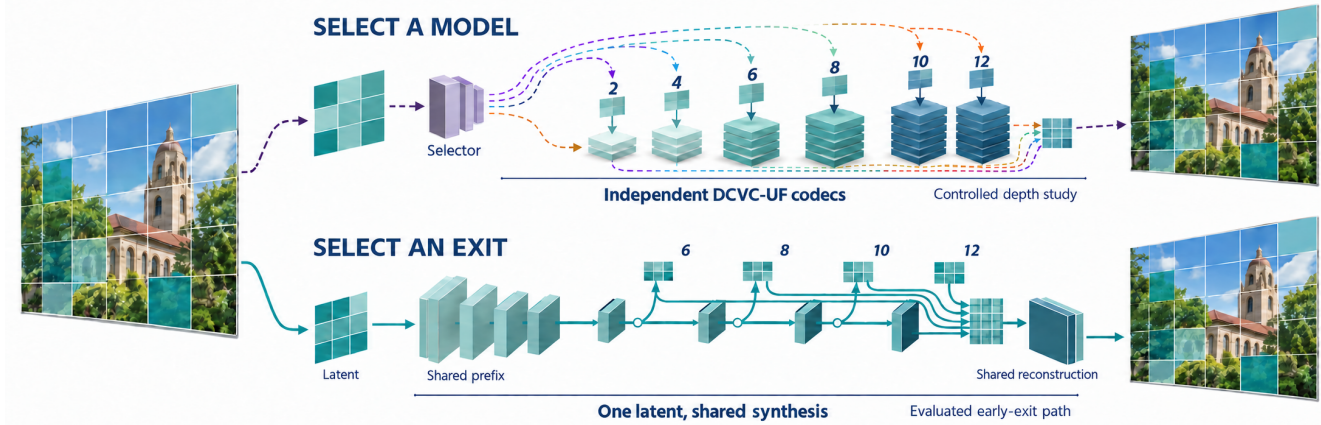


Figure 1. **One allocation principle, two ways to vary synthesis depth.** *Top:* select a complete DCVC-UF codec before encoding a patch; the six-depth bank and its selector form the controlled, prospective study. *Bottom:* select a stopping depth within one shared-latent decoder; this is the evaluated early-exit path. The paths are alternatives, not successive stages. AI-generated conceptual illustration: picture planes are illustrative, not dataset samples or reconstructed outputs. Exact execution and measured allocations appear in Figs. 2 and 6.

sequences; its paired routing margin remains statistically unresolved. This separates evidence for useful allocation from evidence that a source-independent stopping control reliably captures that value.

Our contribution has two parts. First, we provide a spatially conditional DCVC-UF synthesis path with a shared latent, reusable prefixes and feature-aligned exits. Second, we evaluate the value of learned placement against uniform depth and blind mixtures, then test how that conclusion changes under a common measured-loss cap and fixed controls. This connects the architecture to a falsifiable efficiency claim: routing should earn its additional cost through better allocation, rather than through the availability of shallow exits alone. Frozen-weight interventions and a separate-stream decoding check probe the mechanism. The independent-depth study remains ongoing; the present evidence establishes synthesis arithmetic savings, while complete codec speedups require measured execution.

2. Related Work

Making learned decoding cheaper. Shallow synthesis transforms reduce decoder computation globally; stronger analysis or iterative encoding can compensate for some of the lost reconstruction capacity [12]. DCVC-RT instead emphasises operational costs such as memory traffic and function calls [4], while DCVC-UF increases video throughput with chunk-based coding and streamlined entropy interaction [6]. We modify only DCVC-UF’s intra synthesis transform. These approaches motivate two controls in our study: a fixed shallow decoder isolates the value of adaptation, and full execution measurements are required before arithmetic reduction can be called acceleration.

Choosing a network or choosing an exit. ClassSR learns to assign super-resolution patches to networks of different capacity [5]. APE instead lets patches leave a multi-exit network according to predicted incremental reconstruction benefit [9]; it already shares the reconstruction tail across exits. RBQE uses early termination for quality enhancement of compressed images [10]. Swift also introduces compute-dependent early exits in a layered neural video decoder, with different output resolutions [2]. FLEX-UF addresses the codec-specific case within DCVC-UF synthesis at fixed output resolution: one entropy-decoded latent feeds conditional feature tiles, exit-specific adapters align those features, and reconstruction occurs after full-frame assembly. Our proposed independent-depth bank follows the ClassSR allocation principle, but selecting a codec also selects an analysis transform and entropy model.

Adaptive capacity inside a codec. Slimmable compressive autoencoders jointly optimise rate and distortion at multiple network widths [11]. AdaNIC is the closest spatial-capacity precedent: blockwise spatially adaptive operators and a router vary transform width within an image [8]. For video, cgSlimDecoder learns channel allocations across decoder modules under complexity constraints [3]. FLEX-UF varies *depth* through nested, constant-width synthesis blocks and executes later stages on a shrinking tile batch. Its evaluation asks an additional question: how much of the saving requires content-aware placement after blind mixtures have already reduced the average depth?

Spatial selection among complete codecs. Blard et al. train content-specialised codecs of the same architecture, search their regional rate–distortion costs, and signal a mode

map [1]. Their method also processes connected regions of one mode jointly. Our region-merging diagnostic therefore tests an existing strategy in DCVC-UF, rather than introducing it. The planned bank varies synthesis depth and investigates selection before candidate encoding; the evaluated shared-exit path instead retains one representation. These distinctions separate specialisation, capacity and representation reuse, which cannot be inferred from a comparison of model sizes alone.

3. Spatially Adaptive DCVC-UF

3.1. The allocation problem

Let $\mathbf{a} = (a_1, \dots, a_N)$ assign a synthesis option to each region. We seek the least computation that keeps the assembled reconstruction within a quality-loss budget:

$$\min_{\mathbf{a}} \mathcal{C}(\mathbf{a}) \quad \text{subject to} \quad \Delta(x, \hat{x}_{\mathbf{a}}; \hat{x}_{\text{ref}}) \leq \delta, \quad (1)$$

where $\mathcal{C}$ is the chosen compute cost and $\hat{x}_{\text{ref}}$ fixes the reconstruction anchor. The main sweep uses $\Delta_{444} = 10 \log_{10}[D_{444}(x, \hat{x}_{\mathbf{a}})/D_{444}(x, \hat{x}_{\text{ref}})]$, with $D_{444} = \|x - \hat{x}\|_2^2/(3HW)$ on equal-channel, unclipped YCbCr 4:4:4 tensors. Positive values denote a quality loss in dB. The constraint belongs to the *assembled image*: spatial interactions can invalidate an additive tile-error approximation.

For early exit, a_t identifies a stopping point after one latent has been recovered. For the prospective model bank, a_t identifies a complete codec before encoding, making both payload rate and reconstruction depend on the assignment. The bank must therefore also be compared at matched rate, including expert signalling (Sec. 5). We implement shared exits below.

3.2. Shared latent, variable depth

Let $\hat{y}$ be the recovered latent of the fixed DCVC-UF analysis transform and entropy model. The synthesis network upsamples it by two and processes 384-channel features at one eighth of the image resolution. Four of its twelve trunk blocks form the full-frame stem; the remaining eight form four conditional block pairs.

The stem output is partitioned into 32×32 feature tiles, each covering 256×256 image pixels. For a tile assigned depth k , let $f_t^{(k)}$ be its feature after the corresponding block pair and let A_k be its adapter. The complete image is reconstructed as

$$\hat{x}_{\mathbf{k}} = H\left(q \odot G\left[\text{stitch}_t A_{k_t}(f_t^{(k_t)})\right]\right), \quad (2)$$

where G is grid-conditioned repair, q is the decoder's quality scaling, and H is the common output head, including pixel shuffle. Adaptation acts on features before the shared head, so tiles with different stopping depths need not be stitched directly in image space.

Reachable depth and actual execution. The reachable depths are $\mathcal{K} = \{6, 8, 10, 12\}$; depths two and four are inside the shared stem and are not evaluated exits. At each conditional stage the implementation gathers only the surviving tiles, executes the next block pair, and writes departing tiles through their adapters. A deeper tile reuses all computation already performed. Evaluating all four candidates and masking their reconstructions would not implement this execution saving. For the recorded router map at 0.1 dB, the active population decreases from 40 to 27, seven and three tiles (Fig. 3).

3.3. Exit adaptation and boundary consistency

An intermediate feature is not automatically compatible with the final head. The exit adapter supplies the missing transformation with pointwise residual operators. Its residual output is zero-initialised, preserving the inherited feature at initialisation. Adapter capacity scales with the depth replaced. Pointwise operators avoid adding an exit-specific spatial receptive field; their arithmetic is included in the cost model. The deepest exit uses an identity connection rather than an additional learned adapter.

Independent tile execution still removes context from the trunk's spatial operators. The conditional trunk uses replicate padding without a halo. After stitching, a depthwise spatial filter and pointwise mixing act on the full feature canvas. A learned gate indexed by position within the tile gives the repair access to the known boundary lattice. The head then operates on the complete canvas. This design places context recovery at the output stage rather than expanding every conditional block with an overlapping halo. We measure the trained checkpoint's dependence on adapters and repair in Sec. 4.7; matched retraining is needed to choose their capacity.

3.4. Learning exits and their selector

The shared-latent model is initialised from released DCVC-UF weights. Analysis and entropy parameters remain frozen; the synthesis network, adapters and repair are fine-tuned. Training samples tile depths and applies reconstruction supervision through the mixed tiled path. Thus the adapters learn under the same stitching and full-frame head used at inference. The inspected training configuration also enables a full-depth reconstruction anchor against the frozen release and adjacent-exit feature distillation. Its objective can be written as

$$\mathcal{L} = \mathbb{E}_{\mathbf{k}}[R + \lambda D(x, \hat{x}_{\mathbf{k}})] + a\mathcal{L}_{\text{anchor}} + b\mathcal{L}_{\text{feature}}. \quad (3)$$

Here the latent-rate term R is shared across exits. With the front end frozen it does not optimise exit-dependent rate. The archived fine-tuning configuration uses 512-pixel crops and batch size eight; it is distinct from the new full 105-

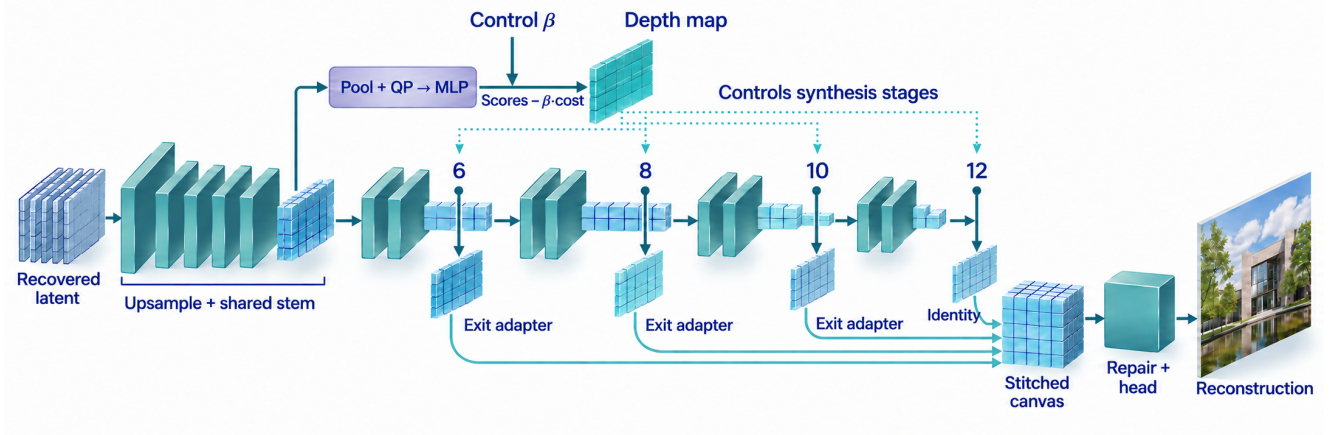


Figure 2. **Share the representation; spend further depth only on surviving tiles.** A four-block shared stem supplies the MLP with features and QP. Cost-adjusted scores select exits at total depths 6/8/10/12. Each departing feature passes through its adapter (identity at depth 12) into one canvas, followed by full-frame repair and reconstruction. The illustration follows the implemented connectivity; tensor contents and the output scene are conceptual, AI-generated artwork rather than measured reconstructions.

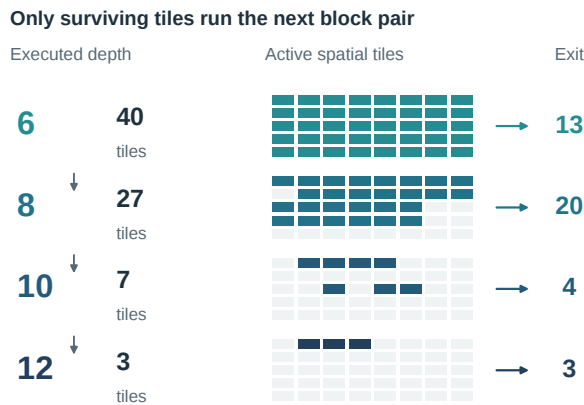


Figure 3. **A spatial decision becomes a smaller workload.** The recorded videoSRC05 map at QP32 leaves fewer tiles active at each successive block pair. Coloured cells run the indicated stage at their padded spatial coordinates; pale cells have already exited. Departures feed the aligned canvas in Fig. 2. The suffix evaluates 77 tile pairs versus 160 for an all-deep map; this work count does not measure GPU time.

epoch official-recipe depth study in Sec. 5. The supplement records the available training provenance and its limits.

The stopping predictor projects the stem from 384 to 48 channels, pools each tile’s mean and standard deviation, and incorporates normalised QP. An MLP with two 256-unit hidden layers produces exit log scores. For candidate tile errors $M_{t,k}$ and normalised costs c_k , its soft training labels favour low values of $M_{t,k} + \lambda_c c_k$. Soft-label cross-entropy and a regret term penalise costly decisions while allowing nearly equivalent exits to receive similar scores.

The checkpoint uses $\lambda_c = 1.3 \times 10^{-5}$ and temperature ratio 0.1; the exact target, input layout and optimisation settings are given in the supplement. This objective learns a preference among exits, not a bound on final image error.

3.5. Stopping under a quality target

For equal padded tiles, the arithmetic model is

$$C(\mathbf{k}) = \frac{1}{N} \sum_t c_{k_t}, \quad S(\mathbf{k}) = 100[1 - C(\mathbf{k})]. \quad (4)$$

Shared stem and output work, exit adapters and conditional depth are included in c_k . Entropy-recovery networks, arithmetic coding, router execution, signalling and system overhead are outside this synthesis MAC quantity. The four costs are approximately $\{0.609, 0.758, 0.870, 1.010\}$ relative to full-frame synthesis. In particular, a tiled all-deep map can cost more than the full-frame reference because repair is still executed.

Source-informed search minimises $M_{t,k} + \lambda_c c_k$ per tile and adjusts λ for a target table loss. The learned alternative uses

$$k_t(\beta) = \arg \max_k \{\ell_{t,k} - \beta c_k\}, \quad (5)$$

where ℓ denotes log scores. Uniform depth selects one exit per frame. Bayer dithering mixes adjacent exits with a fixed spatial ordering. Its mixture fraction is calibrated under the same target, so it controls for finer average depth without predicting spatial difficulty.

In the recorded sweep, all controls are calibrated per source frame using stored error tables. This measures the allocation available with encoder-side source access. The fixed-control experiment in Sec. 4.6 removes per-source calibration. Increasing β cannot increase selected cost, but

source distortion need not be monotone; the supplement audits the finite assignment path for both router and dither.

3.6. Quality references and decision information

The table target compares padded YCbCr 4:4:4 candidate errors to a released-weight full-depth reference. Final reported loss instead compares the cropped mixed output to the cropped, fine-tuned e15 full-frame output. These anchors are not interchangeable: the encoder/entropy weights match, while inherited synthesis weights have changed. Full-frame repair also couples neighbouring tiles. A table threshold therefore does not guarantee the same numerical loss after final reconstruction. We retain this distinction in the nominal sweep and add a retrospective common-reference delivered-loss comparison. A separate QP32 replay on all 53 frames measures the e15 full-frame RGB loss against released synthesis as 0.00908 dB; details are in the supplement.

An encoder-selected map must be signalled to the decoder; alternatively, a decoder-local predictor must recover the same decisions from available features and a reproducible control. We use explicit two-bit exit indices in a CPU stream-only check (Sec. 4.7). Its common payload confirms the representation-sharing path, while headers and map bytes remain part of the complete rate. Source access is legitimate at the encoder; its candidate evaluations and calibration must be charged to the decision cost.

4. Experiments

4.1. Evaluation protocol

We evaluate the first intra frame of 53 video sequences at QP values $\{0, 16, 32, 48, 63\}$, giving 265 frame-QP pairs. Every policy uses the same pinned e15 shared-latent checkpoint, the same tile geometry and the same per-exit cost vector. The learned rule uses one fixed router checkpoint. Six target budgets span 0.05–0.5 dB. The source-calibrated archive contains actual mixed reconstructions. Additional CPU experiments test fixed controls, the released reference and component dependence. Every numerical decoder comparison concerns DCVC-UF; no experiment here measures complete GPU codec latency.

Two complementary protocols answer different questions. The *nominal sweep* compares maps selected under the same table target, retaining the common feasible cases of all four rules: 204 pairs at 0.05 dB, 263 at 0.1 dB and 265 at each larger target. The *delivered-cap analysis* reselects from the outputs already measured at those targets under a common cropped YCbCr 4:4:4-loss cap, with the full-frame e15 output as an explicit fallback. It retains all 265 pairs at every cap. The second protocol is retrospective and source-aware; its cost omits acquiring and rejecting candidates.

Confidence intervals use 5,000 paired bootstrap draws over sequences, keeping QPs from the same sequence to-

gether. They quantify sequence-sample variation conditional on fixed weights and controls, without including training or calibration uncertainty. The supplement also reports a conservative content-group sensitivity check. A separate fixed-cohort comparison retains the same 204 pairs across the entire nominal sweep, avoiding changes of sample membership as the target relaxes.

4.2. The reconstruction value of depth

The stored uniform-exit reconstructions provide a direct depth control. Across all 265 pairs, their mean padded YCbCr 4:4:4 losses against the released full-frame reference are 0.205/0.097/0.061/0.038 dB at 6/8/10/12 blocks. Their modelled MAC savings are 39.12/24.22/13.03/—0.95%, respectively. Additional depth has diminishing average benefit and non-uniform local benefit: mean within-frame tile-gain IQRs are 0.075/0.034/0.024 dB for successive block pairs. The supplement gives the distributions and paired intervals. These measurements concern actual uniform maps on padded YCbCr 4:4:4; they motivate spatial allocation but are not cropped mixed-map losses or results for independently trained shallow codecs.

4.3. Early exit reduces work; routing improves its allocation

At the nominal 0.1 dB target, FLEX-UF with learned routing saves 27.97% of modelled synthesis MACs. Source-informed search reaches 29.52%, while uniform depth and dithering reach 21.07% and 25.43%, respectively (Fig. 4). The ordering separates three effects. Shallower synthesis supplies most of the reduction. Mixing adjacent depths improves the available cost resolution. Content-dependent placement contributes a further 2.54 percentage points over dithering, with a paired 95% interval of [1.83,3.26].

4.4. The gain persists under a common delivered-loss cap

Does the margin persist when every selected output meets the same measured loss cap? For each policy and frame-QP pair, we deduplicate the maps measured at the six nominal targets, retain those meeting the cropped Δ_{444} cap within 10^{-4} dB, and choose the cheapest. Repeated maps have identical recorded losses and costs. The evaluated e15 full-frame output supplies a zero-loss, zero-saving fallback; no quality point is interpolated.

Table 1 reports the 0.1 dB case. The router saves 27.75% versus 24.81% for dithering, a 2.93-point paired difference [2.11,3.79]. Their mean achieved losses are also close: 0.0786 and 0.0791 dB. Only one router case uses the full-frame fallback, compared with two dither cases. The margin survives a common measured cap. Selection remains retrospective and source-aware, omits candidate-acquisition cost, and searches only the recorded outputs.

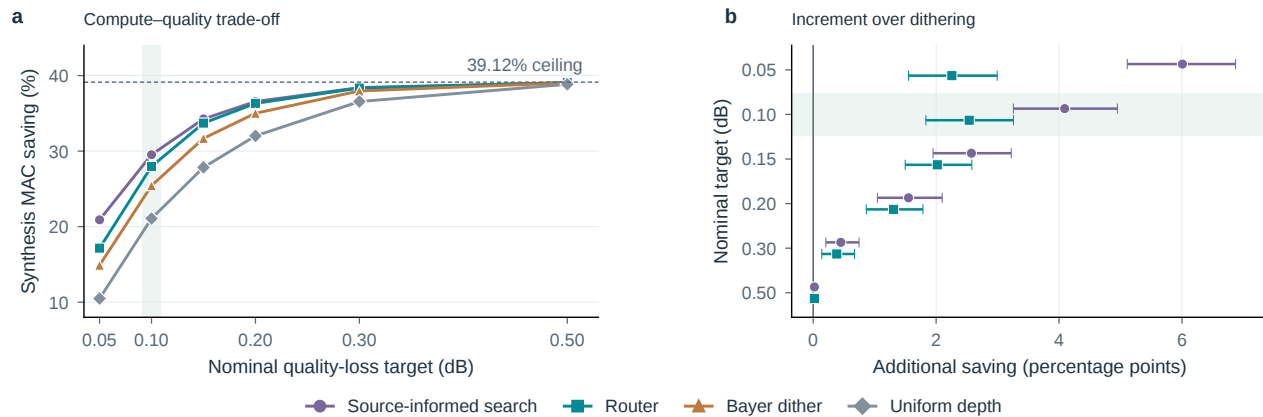


Figure 4. **Early exit supplies the saving; spatial routing adds a budget-dependent margin.** **a**, Modelled synthesis MAC reduction for four stopping rules. **b**, Paired advantages over Bayer dithering, with 95% sequence-cluster intervals; the shaded row highlights 0.1 dB. QP and fixed-cohort analyses are in the supplement. All curves use per-source calibration. MAC savings exclude router and signalling overhead and are not measured runtime reductions. Targets use the Δ_{444} MSE-ratio loss defined in Sec. 3.

Table 1. **Recorded outputs under a common 0.1 dB YCbCr 4:4:4-loss cap.** All 265 pairs are included. Loss is relative to cropped e15 full-frame synthesis. MAC saving excludes the retrospective candidate-selection cost.

Policy	MAC saved (%)	Mean loss (dB)	Fallbacks (/265)
Uniform depth	22.98	0.0670	11
Bayer dither	24.81	0.0791	2
Learned router	27.75	0.0786	1
Source search	29.19	0.0779	2

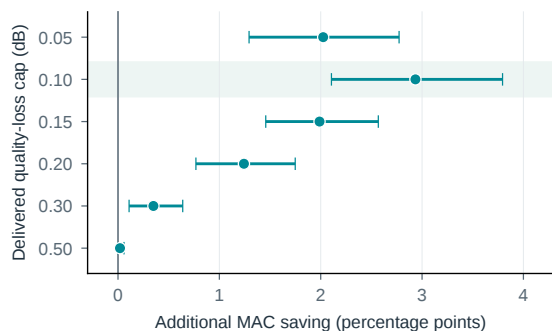


Figure 5. **Placement matters most before the cheap exit saturates.** Paired router–dither MAC-saving differences under a common measured YCbCr 4:4:4-loss cap, with 95% sequence-cluster intervals. All 265 pairs remain; e15 full-frame is the fallback. Selection is retrospective and source-aware.

4.5. Why the routing margin contracts

The cheapest reachable depth imposes a 39.12% ceiling on the arithmetic model. As the target relaxes, more tiles can take that exit and there are fewer consequential spatial decisions. The nominal router–dither margin declines from 2.54 points at 0.1 dB to 1.31 at 0.2 dB, 0.38 at 0.3 dB and 0.02 at 0.5 dB. Under the delivered-cap protocol the margin similarly falls from 2.93 to 0.35 points between 0.1 and 0.3 dB. The fixed-cohort result and QP breakdown in the supplement indicate that this behaviour is not explained solely by a changing feasible subset. At the 0.1 dB delivered cap, routing saves more in 170 pairs, less in 24, and ties in 71. At 0.5 dB, 262 of 265 pairs tie; the residual gain comes from one sequence.

The maps in Fig. 6 make the mechanism visible. At the 0.1 dB target, the router spends different depths on different regions. At 0.3 dB, search, routing and dithering all assign depth six on this frame, saving 39.1% MACs at 0.201 dB measured YCbCr 4:4:4 loss. We select this example from the five archived thumbnails by proximity to the aggregate 0.1 dB margin. Source luma supplies context, not a visual quality comparison.

A histogram-preserving diagnostic further separates spatial placement from exit proportions. Randomly permuting a map leaves its cost unchanged. We compute the exact expected table MSE under that permutation and compare it with the recorded assignment. At 0.1 dB, the router’s mean placement gain is 0.0213 dB [0.0167, 0.0262], whereas dithering’s is 0.0007 dB [−0.0002, 0.0017]. This table-based association holds the histogram fixed; replaying permuted maps through the final head remains a separate ablation.

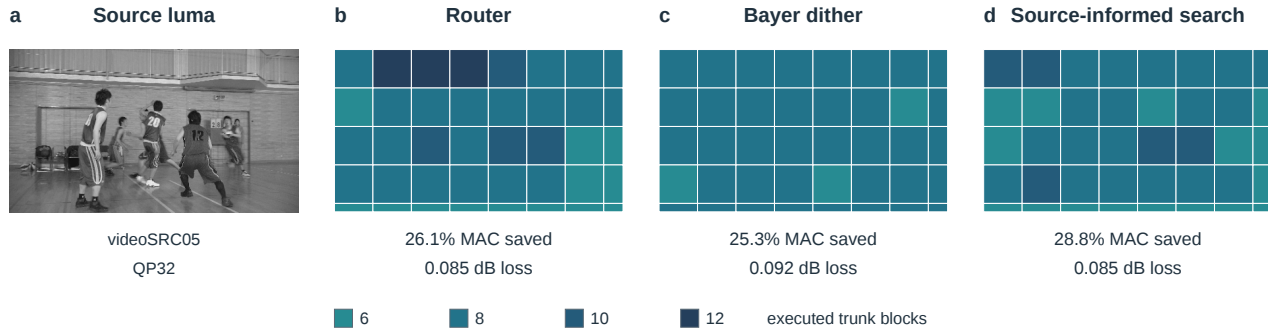


Figure 6. **Adaptive depth changes where synthesis work is spent.** **a**, Source luma for spatial context. **b–d**, Recorded router, Bayer and source-search maps at the 0.1 dB target, QP32. Colour gives exit depth; annotations give synthesis MAC saving and cropped YCbCr 4:4:4 loss. Uniform depth eight gives 24.2% saving and 0.082 dB loss. Maps use valid image extent; the text states the example-selection rule.

Table 2. **Fixed controls trade average saving against quality risk.** Actual CPU reconstructions on all 53 QP32 frames. Controls are fitted to padded 444 loss on other sequences; the table reports cropped RGB loss against e15 full-frame. Equal calibration targets do not imply equal achieved quality. The final column counts losses above 0.1 dB.

Control	Policy	MAC saved (%)	RGB loss (dB)	> 0.1 (/53)
Mean	Uniform	19.78	0.0775	14
	Dither	23.80	0.0881	16
	Router	25.68	0.0868	17
Q90	Uniform	13.03	0.0516	2
	Dither	14.20	0.0564	2
	Router	16.74	0.0568	4

4.6. Fixed controls test the routing advantage

Source-calibrated allocation gives the selector information that a cheap stopping rule would ideally avoid. We therefore fit its scalar control on four sequence folds and apply it to the fifth, keeping all QPs of each sequence together. Calibration minimises compute subject to a mean or 90th-percentile (Q90) table-loss target. We then replay every fixed map at QP32 through the actual network: 53 frames, three policies and two controls give 318 reconstructions, with no quality-based fallback.

Table 2 reports cropped RGB loss against e15 full-frame synthesis. Q90 calibration lowers the observed frequency of losses above 0.1 dB, at the cost of less arithmetic saving. The router–dither saving difference is 1.87 points under mean control, with a paired 95% interval of $[-1.60, 5.19]$, and 2.54 points under Q90, with interval $[-0.42, 5.54]$. Both include zero, and the policies do not have exactly matched achieved quality. Thus the source-calibrated margin has not yet been established for fixed controls.

This is a calibration diagnostic on development-used data, not an untouched test set. Two similar-content pairs cross the sequence folds; grouping them in a bootstrap sensitivity check leaves both intervals crossing zero. An aligned released-weight replay also measures the reference drift: e15 full-frame loses 0.00908 dB mean RGB PSNR against released D12. Using released D12 raises absolute policy losses by that same per-image offset without changing their paired differences. The supplement gives the split, predictor and metric audits.

4.7. Checking the mechanism

Do exits need learned alignment? With the checkpoint and 53 Q90 maps fixed, we replace adapters, repair, or both by identity and retain all images. Adapter removal costs 1.2309 dB mean RGB PSNR, with a paired 95% interval of $[0.8765, 1.6582]$. Repair removal costs only 0.00208 dB $[0.00060, 0.00363]$; removing both costs 1.2538 dB. The trained model depends strongly on exit alignment. The small repair benefit motivates a matched-retraining and timing comparison before increasing its capacity. These interventions measure dependence of fixed weights, not the best performance attainable without each module.

Can the decoder operate from a stream alone? A separate CPU decoder receives the common entropy payload and explicit two-bit exit indices. On two predeclared QP32 frames with four maps each, all eight images, latents, symbols and entropy indexes match encoder-side reconstruction exactly; inner entropy bytes stay identical across maps. This verifies the shared-payload path on those cases. Map/header bytes and malformed-stream checks are in the supplement; native CUDA interoperability and full-cohort timing remain unmeasured.

4.8. What the encoder pays for a stopping decision

Source-informed search needs candidate reconstructions to populate its error table. The archived QP32 benchmark records 1,023.26 ms for four distinct candidate outputs and 336.87 ms for a shared tiled pass. These upper medians pool 24 trials over eight frames; raw timing distributions and device load were not retained. They indicate the cost of acquiring decision information, not an end-to-end speedup. The supplement details the stages.

A fixed-control predictor can avoid this table, but the evaluated source-calibrated router cannot: its per-frame β uses the same error information. The next latency experiment must compare full decision paths, including feature generation, calibration, signalling, conditional execution and final verification. At a loose quality budget, where the router and blind maps nearly coincide, even small additional overhead may remove the practical benefit of content prediction.

5. From Shared Exits to Independent Depths

Shared early exit commits every region to one encoded representation. An independent codec can specialise its representation for a smaller synthesis network, but forfeits that reuse. This motivates our complementary 2/4/6/8/10/12-block DCVC-UF bank (Fig. 1), where a later ClassSR-inspired selector will choose a codec before encoding each region. D2/D4/D6 are training; D8/D10 are planned and D12 is the released anchor. No six-expert routing result is claimed.

A controlled capacity frontier. The scratch runs preserve the pinned official 105-epoch recipe [7], including all 384,795 Open Images train_0/1/2 images, batch size 16 and FP32. Analysis and entropy weights learn alongside synthesis, so equal QP does not imply equal coded rate. A frozen epoch-20 diagnostic on 100 DIV2K centre crops uses 2,000 actual CPU bitstreams across D2/D4/D6 and released D12. At 0.2 payload bpp, D6 exceeds D2 by 0.183 dB RGB on the 99 images with common rate support. These interim curves establish a measurable depth frontier; final same-recipe D12 and matched shared-exit training are still required to attribute differences to depth or weight sharing. The supplement gives the schedule, counts and complete curves.

The spatial cost of specialisation. Patchwise expert selection also changes geometry and entropy boundaries. With identical D6 weights, independently coded regions lose quality at matched rate; increasing context alone does not remove their rate cost. In a separate fixed D2/D6 assignment, merging compatible neighbouring regions improves matched-rate quality while retaining the same per-pixel depth map. These controls, detailed in the supplement, make patchwise D12

and blind expert mixtures essential baselines before a selector receives credit. The eventual comparison must match achieved rate and quality, charge signalling and grouping, and measure the effect of expert occupancy on latency. It will test when independent specialisation is worth more than shared-prefix reuse.

6. Discussion

Adaptation should pay for the decisions it changes. FLEX-UF identifies a budget-dependent role for spatial allocation. Under source calibration, tight quality targets leave room for useful placement. Near the shallowest-exit ceiling, a blind mixture reaches nearly the same arithmetic saving. This motivates a budget-conditioned choice between fixed, blind and learned policies, with its switching rule fixed on validation data.

Reuse and specialisation are different commitments. Shared exits retain one representation and the work already performed; independent experts can adapt their encoders and synthesis capacity more freely. The bank adds model residency, entropy boundaries and occupancy-dependent execution costs. Its quality must be paired with those costs, not with shared-prefix arithmetic. Multiple exits inside each expert are outside the present design.

The remaining evidence gap. The source-calibrated study measures available allocation value, while the fixed-control replay leaves the routing increment unresolved. Content-grouped calibration on untouched images is the next test of that increment. Native bitstreams and paired wall-clock measurements must then include entropy work, prediction, tile movement, grouping and fallback. The current confidence intervals cover sample variation at fixed weights, not independent training seeds. Results are confined to intra reconstruction: YCbCr-MSE and RGB diagnostics do not establish perceptual equivalence, temporal consistency or performance of DCVC-UF's chunk-based video path.

7. Conclusion

FLEX-UF makes synthesis depth a spatial decision inside DCVC-UF. A common latent and prefix support aligned early exits, while a shrinking tile batch turns stopping decisions into omitted block evaluations. The measured arithmetic savings have two sources: cheaper average depth and a smaller, budget-dependent increment from learned placement. Blind mixtures and fixed-control replay show why these must be evaluated separately. The independent-depth study extends the allocation question to codec selection; matched training and complete execution measurements will determine when specialisation justifies giving up shared computation.

References

- [1] Théophile Blard, Pierrick Philippe, Théo Ladune, Xiaoran Jiang, and Olivier Déforges. Spatial competition for low-complexity learned image compression. In *IEEE International Conference on Image Processing (ICIP)*, 2026. Accepted; author manuscript arXiv:2605.13243. 3
- [2] Mallesh Dasari, Kumara Kahatapitiya, Samir R. Das, Aruna Balasubramanian, and Dimitris Samaras. Swift: Adaptive video streaming with layered neural codecs. In *19th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, pages 103–118. USENIX Association, 2022. 2
- [3] Zhihao Hu and Dong Xu. Complexity-guided slimmable decoder for efficient deep video compression. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14358–14367, 2023. 2
- [4] Zhaoyang Jia, Bin Li, Jiahao Li, Wenxuan Xie, Linfeng Qi, Houqiang Li, and Yan Lu. Towards Practical Real-Time Neural Video Compression. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12543–12552, 2025. 2
- [5] Xiangtao Kong, Hengyuan Zhao, Yu Qiao, and Chao Dong. ClassSR: A General Framework to Accelerate Super-Resolution Networks by Data Characteristic. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12016–12025, 2021. 1, 2
- [6] Jiahao Li, Wenxuan Xie, Zhaoyang Jia, Bin Li, Zongyu Guo, Xiaoyi Zhang, and Yan Lu. Ultra-Fast Neural Video Compression. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 41311–41321, 2026. 1, 2
- [7] Microsoft. DCVC-UF Image Training Recipe. Source repository, commit cbdae87a5445114cdc7f48816da63ea80bdeac40, 2026. 8
- [8] Lvfang Tao, Wei Gao, Ge Li, and Chenhao Zhang. Adanic: Towards practical neural image compression via dynamic transform routing. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16879–16888, 2023. 2
- [9] Shizun Wang, Jiaming Liu, Kaixin Chen, Xiaoqi Li, Ming Lu, and Yandong Guo. Adaptive Patch Exiting for Scalable Single Image Super-Resolution. In *European Conference on Computer Vision (ECCV)*, 2022. 1, 2
- [10] Qunliang Xing, Mai Xu, Tianyi Li, and Zhenyu Guan. Early exit or not: Resource-efficient blind quality enhancement for compressed images. In *European Conference on Computer Vision (ECCV)*, 2020. 2
- [11] Fei Yang, Luis Herranz, Yongmei Cheng, and Mikhail G. Mozerov. Slimmable Compressive Autoencoders for Practical Neural Image Compression. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 2
- [12] Yibo Yang and Stephan Mandt. Computationally-Efficient Neural Image Compression with Shallow Decoders. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023. 2